MIL-C-43712C
22 December 1986
SUPERSEDING
MIL-C-43712B
28 December 1979

MILITARY SPECIFICATION

# CANS, WATER, MILITARY, PLASTIC, COLLAPSIBLE

This specification is approved for use by all Department and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers 5 gallon collapsible, plastic water cans and closures.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| L-P-390 | - Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density) |
| PPP-B-601 | - Boxes, Wood, Cleated Plywood |
| PPP-B-636 | - Boxes, Shipping, Fiberboard |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7240

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (CFR: 1-199)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Standard product. The water cans delivered under this document shall be the manufacturer's commercial product, except for changes necessary to comply with requirements of this document.

3.3 Federal Food and Drug Act certification. Prior to the approval of the first article, if one is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the plastics and other materials proposed to be used in the water cans, caps, and cap liners, as well as all antioxidants and other substances incorporated into these materials meet the requirements of 21 CFR - Food and Drugs: Indirect Food Additives: Polymers, Subpart B, Part 177, Chapter I, Title 21 Code for Federal Requirements for Contact with Food. Acceptable evidence of meeting the requirements of this document shall be a certified test report from a recognized independent testing laboratory acceptable to the Department of the Army Surgeon General.

3.4 Materials and components. Material and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for his water can, provided the completed item complies with all provisions of this document.

3.4.1 Can bodies. Can bodies shall be molded from low density polyethylene molding material conforming to type I, class L, grade 2 of L-P-390, as specified herein. The color of the material shall be Olive Drab 34087 (lusterless) of FED-STD-595.

3.4.2 Can caps. Can caps shall be molded from high density polyethylene molding material conforming to type I, class H, grade 2, 4, or 5 of L-P-390. The color of the material shall be Olive Drab 34087 (lusterless) of FED-STD-595.



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* 3.4.3 Cap liners. Cap liners shall be made from neoprene rubber or a suitable plastic material.

3.5 Design and construction. The collapsible water cans shall be designed as specified herein.

3.5.1 Construction methods and dimensions. The body of the water can shall be fabricated from material specified in 3.4.1, and shall consist of bottom, sidewalls, and neck that are formed in one piece by blow-molding. The bottom flat surface shall be minimum 0.025-inch thickness. The bottom flat surface is defined to be the bottom portion of the body that is flat when placed in the normal upright position on a horizontal surface. The thickness of the sidewalls shall be a minimum of 0.012 inch. The sidewall of the body is the area from the bottom of the neck to the bottom flat surface. The thickness at the neck (thread area) shall be a minimum of 0.080 inch. The body shall have a capacity of not less than 5 US gallons and shall have dimensions of 9 1/4 ± 1 1/2 inches long, 9 1/4 ± 1 1/2 inches wide, and 13 1/2 ± 1 1/2 inches high.

3.5.2 Stability. The water can shall remain upright when filled to capacity and tested in accordance with 4.5.1. The unfilled water can shall collapse to a maximum height of 3 1/4 inches. The water can shall not fracture when tested in accordance with 4.5.3. The threads on the neck shall be a minimum 60 mm buttress type, fully formed and free from flash or thread misalignment at the parting line or other imperfections. The sealing face of the threaded area of the neck that contacts the liner of the cap shall be smooth, flat, and free from any imperfection that prevents the sealing of the cap. The sealing face may have an out-of-roundness in the plane parallel to the sealing face, provided that, when the cap is tightened, there shall be no leakage of water. The neck area shall show no evidence of warpage. There shall be no leakage of water between the neck and the cap joint when tested in accordance with 4.5.1.

3.5.3 Cap. The cap and cap retaining strap shall be molded from material specified in 3.4.2 and may be molded in one or two pieces. The cap shall have threads mating with the neck thread. The cap shall have a liner that fits snugly in the cap. The liner shall be made from material specified in 3.4.3. The cap retaining strap shall be a minimum of 1/2 inch wide by a minimum of 1/32 inch thick. The cap retaining strap shall hold the cap captive to the water can. The cap shall rotate 360° without the cap retaining strap twisting.



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3.5.4 Handle. The handle shall be made from material specified in 3.4.1 and shall be the design normally used by the manufacturer. The handle shall be attached to the body or molded as part of the body. The handle shall not protrude above the top of the cap when the cap is in its lowest position. When the water can is filled with water, the handle shall not break or separate from the water can when tested as specified in 4.5.2.

* 3.5.5 Resistance to rough handling. The water can shall not fracture when tested as specified in 4.5.3.

3.6 Marking. Each water can cap and bottom shall be permanently marked with the manufacturer's identification in accordance with the manufacturer's standard practice. The top of the water can cap shall be permanently marked "Potable Water Only" in characters a minimum of 1/4 inch in height and raised at least 0.007 inch high by an embossed molded-in process.

3.7 Workmanship. The water cans shall be clean and free from flash, bubbles, cuts, tears, holes, breaks, cracks, pinholes, pits, ripples, warpage, blisters, scratches, or any weld or sink marks. The water can shall be free from dust, dirt, and foreign matter embedded in the polyethylene.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process to assure compliance with all dimensional requirements.

* 4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), three sample water cans shall be examined for the defects specified in 4.4.2 and 4.4.3. Testing shall be performed on one can as specified in 4.5.1 and in 4.5.2, and three cans shall be tested as specified in 4.5.3. The presence of any defects in table I, any dimension not within specified requirements, or the failure of any unit to pass any of the applicable tests shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105. Quality conformance inspection shall consist of all examinations and tests in 4.4 and 4.5.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of water cans. The sample unit shall be one water can. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total defects.



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TABLE I. End item visual defects

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Finish | Color not as specified | | 201 |
| Construction workmanship | Not free of dust, dirt, or foreign matter imbedded in polyethylene | 101 | |
| | Part missing | 102 | |
| | Threads incorrect type | 103 | |
| | Not clean; not free from flash, bubbles, cuts, tears, holes, burns, breaks, cracks, pinholes, pits, ripples, warpage, blisters, scratches, or any weld or sink marks | 104 | |
| Handle | Not attached as specified | | 202 |
| | Protrudes above top of cap when in lowest position | | 203 |
| Cap | Not held captive as specified | 105 | |
| | Thread too tight on neck | 106 | |
| | Liner not as specified | 107 | |
| | Liner loose in cap | | 204 |
| Assembly | Mating parts do not function | 108 | |
| Marking | Missing, incomplete, not legible, not specified type or size, not in proper location | | 205 |

4.4.3 End item dimensional examination. The end item shall be examined for the dimensions for body length, width, and filled and unfilled height as specified in 3.5.1. Any dimension not within the specified tolerances shall be classified as a defect. The lot size shall be expressed in units of water cans. The sample unit shall be one water can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.



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4.4.4 End item testing. The water cans shall be tested as specified in 4.5.1 and 4.5.2. The lot size shall be expressed in water cans of one size. The sample unit shall be one water can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. When a first article is not required, three units from the initial lot shall be tested as specified in 4.5.3, and any nonconformance with any test requirements shall be cause for rejection of the lot. The test in 4.5.3 need not be repeated during the life of the contract, providing there is no change in material composition and method of construction in subsequent lots.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

| Examine | Defect |
|---|---|
| Marking (exterior and interior) | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |
| Materials | Any component missing or not as specified<br>Any component damaged affecting serviceability |
| Workmanship | Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling<br>Bulged or distorted container |
| Content | Number of water cans per container is more or less than required |

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

| Examine | Defect |
|---|---|
| Finished dimensions | Length, width, or height exceeds specified maximum requirement |



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| Examine | Defect |
|---|---|
| Palletization | Pallet pattern not as specified<br>Interlocking of loads not as specified<br>Load not bonded with required straps as specified |
| Weight | Exceeds maximum load limits |
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |

4.5 Methods of inspection.

4.5.1 Leakage and stability test. The water can shall be filled with 5 gallons of water, and the cap shall be screwed down and tightened. The water can shall be placed in the upright position for not less than 1/2 hour and examined to determine compliance with the stability requirement in 3.5.2. The water can shall be placed on one side for not less than 1/2 hour and examined to determine compliance with the leakage requirement in 3.5.2. The can shall be placed on the opposite side, and the test shall be repeated. Inability of the water can to remain upright or to hold 5 gallons without leakage shall constitute failure of the tests.

4.5.2 Handle test. The water can shall be filled with 5 gallons of water. The cap shall be screwed down and tightened. Twisting of the captive molded cap retaining strap shall constitute failure of the test. The test shall be performed in an ambient temperature of 110° ± 5°F. A web strap equal to the inside width of the handle shall be laced through the handle, and the water can shall be suspended. The water can shall remain hanging for 8 hours and then examined for handle strength to determine compliance with 3.5.2. Inability to meet the requirement for no handle separation or any breakage or leakage shall constitute failure of this test.

* 4.5.3 Drop test. Three collapsed water cans (see 4.3 and 4.4.4), as applicable, shall be selected and placed in a freezer at 0°F for 24 hours. Remove the first sample and fill with 4 3/4 gallons (± 1/8 gallon) with water being at 35° ± 3°F. Screw cap(s) on tightly. Drop can from a height of 5 feet onto a solid concrete horizontal surface. The water can shall be examined to determine compliance with 3.5.2. Inability of the water can to withstand the drop without cracking, breaking, or developing leaks shall constitute failure of this test. Fill the second sample with water at 90° ± 5°F and the third sample with water at 130° ± 5°F and repeat test. Any noncompliance with the specified requirements shall constitute failure of this test.



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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each water can shall be collapsed and neatly folded with the neck exposed and then secured with a minimum 1/8-inch wide rubber band. The caps shall then be loosely installed.

5.1.2 Commercial. The water cans shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Forty-eight water cans, preserved as specified in 5.1, shall be packed in a cleated-plywood wood box conforming to overseas type, style optional, type 2 load of PPP-B-601. Each box shall be closed and strapped in accordance with PPP-B-601.

* 5.2.2 Level B packing. Forty-eight water cans, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V3c, V3s, or V4s of PPP-B-636. Shipping containers having a closure or manufacturer's joint secured with staples shall have the staple area covered with fiberboard pad(s) of the same material as the shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

* 5.2.3 Commercial packing. Water cans, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), water cans packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. The pallet pattern shall be in accordance with appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.



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5.4 Marking. In addition to any special marking required by the contract, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The water cans are intended for the storage of drinking water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
d. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Subject term (key word) listing.

Can, plastic
Water can, collapsible

6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-43712C | Cans, Water, Military, Plastic, Collapsible |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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NATICK OP, 1 Apr 84

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